

Healthy Bite: Digital Menu And Food Ordering System

Author Names: Gunjan Dhaduk, Jisna Saji

Guide Name: Ms. Pintukumari Bera

Christ College, Rajkot

Abstract

The rapid adoption of digital food ordering platforms has revolutionized dining; however, most existing solutions prioritize order dispatch while offering negligible transparency regarding nutritional composition, ingredient customization, calorie loads, and dietary suitability. This lack of dietary awareness contributes directly to lifestyle disorders such as obesity, diabetes, and cardiovascular diseases.

Healthy Bite is a specialized, web-based Digital Restaurant Menu and Nutritional Food Ordering System designed to bridge the gap between dining convenience and health transparency. Engineered with a frictionless QR-code table ordering architecture, the system allows seated diners and takeaway guests to scan table-specific QR tokens, browse food catalogs, inspect ingredient transparency, and review an 8-parameter nutritional profile (calories, protein, carbohydrates, fats, dietary fiber, sugar, sodium, and caffeine) before placing an order. An integrated dynamic computation engine recalculates price and macronutrients in real-time as customers modify dish variants and dietary add-ons.

For restaurant management, the platform incorporates a dedicated multi-role portal featuring instant dish availability control, table and QR token administration, simulated digital payment processing, and a 5-stage live Kitchen Display System (KDS). Developed using a custom 3-Tier MVC architecture in PHP, MySQL, HTML5, CSS3, JavaScript (ES6+), and Bootstrap Icons, Healthy Bite guarantees data integrity, responsive client-side state management, and defensive web security against common vulnerabilities (SQLi, XSS, CSRF). Healthy Bite transforms standard dining operations into an interactive, health-aware digital experience.

Keywords: Digital Restaurant Menu, QR Food Ordering, Nutritional Transparency, Calorie Tracking, PHP MVC, MySQL, Kitchen Display System (KDS), Web Application Security.

1. Introduction

In recent years, digital transformation has deeply reshaped the food service industry. Quick-response (QR) digital menus and contactless ordering systems have evolved from temporary pandemic countermeasures into essential expectations for modern restaurant operations. They drastically reduce guest wait times, eliminate paper printing costs, and streamline front-of-house coordination.

Despite widespread adoption, conventional digital menu platforms focus almost exclusively on transactional speed and sales volume, providing minimal or zero nutritional intelligence. Diners are routinely unaware of hidden sugars, excessive sodium, allergen exposure, or caloric density. This information deficit poses significant risks for fitness enthusiasts, diabetic patients, elderly diners, and individuals with strict dietary regimens.

Healthy Bite resolves this dilemma by unifying contactless table ordering with real-time nutritional disclosure. Unlike generic food ordering portals, Healthy Bite treats nutritional data as a first-class citizen alongside pricing. Every dish features an itemized breakdown of calories, macro/micronutrients, and dietary classification tags (e.g., High-Protein, Keto, Vegan, Gluten-Free, Low-Sugar). Crucially, the platform dynamically computes nutritional shifts as diners customize their meals (such as choosing almond milk over dairy or opting for low-sugar dressings).

Beyond the consumer-facing interface, the system equips restaurant owners and kitchen managers with a centralized operations dashboard. This includes real-time order lifecycle tracking through a 5-column Kitchen Display System (KDS), table occupancy management, QR token generation, catalog customization, and revenue analytics. Healthy Bite establishes a transparent, efficient ecosystem that benefits diners and restaurant operators alike.

2. Problem Statement

Contemporary restaurant ordering systems and online food menus suffer from notable architectural and informational limitations:

- **Absence of Nutritional Transparency:** Traditional menus omit vital macronutrient data (calories, proteins, carbs, fats, sugars, sodium), preventing health-conscious decisions.

- **Static, Inflexible Catalogs:** Paper and basic digital menus cannot recalculate nutrition when customers request ingredient customizations or portion variants.
- **High Customer Onboarding Friction:** Many ordering applications force seated restaurant patrons to download bulky mobile apps or complete tedious account registration forms before ordering.
- **Manual and Fragmented Kitchen Operations:** Traditional order tickets (KOT) result in delayed order fulfillment, miscommunication, and poor status tracking.
- **Lack of Real-Time Menu Availability Control:** Out-of-stock dishes cannot be immediately removed, causing diner disappointment and kitchen delays.
- **Security & Concurrency Vulnerabilities:** Unsecured web forms and exposed order endpoints risk tampering, parameter injection, and cross-table ordering conflicts.

Healthy Bite addresses these challenges through a secure, responsive, web-based platform that combines frictionless QR table ordering, dynamic nutrition recalculation, and synchronized kitchen workflow management.

3. Objectives

The primary objectives of the Healthy Bite system are:

- **Develop a Frictionless QR Ordering Platform:** Enable diners to access the digital menu instantly by scanning dynamic table QR tokens without requiring account registration or mobile app installation.
- **Provide 8-Parameter Nutritional Transparency:** Display verified nutritional facts per dish, including calories (kcal), protein (g), carbohydrates (g), fat (g), fiber (g), sugar (g), sodium (mg), and caffeine (mg).
- **Implement Dynamic Macro Recalculation:** Develop a client-server calculation engine that updates both total price and aggregate nutritional values in real-time as variants and custom add-ons are toggled.
- **Deliver an Interactive Food Catalog:** Support category filtering, search queries, food classification badges (Veg/Non-Veg), and dietary lifestyle tags.

- **Streamline Table and QR Management:** Enable restaurant managers to generate, preview, print, and assign unique secure QR tokens to dining tables with automated occupancy tracking.
- **Implement a Live Kitchen Display System (KDS):** Provide a real-time 5-stage Kanban queue (placed, accepted, preparing, ready, completed) for kitchen staff.
- **Provide Live Customer Order Tracking:** Allow diners to track their order's preparation and fulfillment status via asynchronous status polling.
- **Implement Simulated Multi-Method Settlement:** Facilitate order settlement across simulated Cash, UPI, and Card options with cryptographically secure transaction reference generation.
- **Enforce Defensive Web Architecture:** Ensure application security through PDO parameterized SQL queries, context-aware output escaping (XSS prevention), constant-time CSRF verification, and role-based session middleware.

4. Scope of the System

The system is scoped across three operational stakeholder roles and a focused future development roadmap:

A. Dining Customer (Frictionless Web Access)

- Scan table QR code to resolve verified restaurant, branch, and table identity.
- Browse category-wise healthy meals with visual dish photography and nutritional tags.
- Search and filter menu items based on dietary preferences.
- Open interactive food customization modals to select portion variants and dietary add-ons.
- View real-time reactive updates to item price and nutritional values.
- Manage local cart state (hb_cart) with persistent storage and line-item quantity controls.
- Complete guest checkout by providing contact details and dining preference (Dine-In or Takeaway).

- Settle orders via simulated payment options (Cash, UPI, Card).
- Track live kitchen order progress via real-time HTTP polling.
- Submit customer ratings and qualitative dining reviews.

B. Restaurant Owner & Kitchen Staff (Operations Portal)

- Authenticate securely via role-based access control (Owner / Manager / Staff).
- Monitor high-level business indicators (total revenue, daily order volume, active tables).
- Manage menu catalog: create, edit, and delete food items with base price, images, and nutritional metrics.
- Toggle instant dish availability (In Stock / Sold Out) with live menu reflection.
- Configure variants (e.g., Regular, Large) and add-on customizations with specific price and macro adjustments.
- Administer dining tables, generate dynamic QR codes, download vector QR images, and manage table occupancy.
- Operate the live Kitchen Display System (KDS) Kanban board to advance orders across preparation stages.
- Review customer ratings and publish official restaurant responses.
- Access sales breakdowns and nutrient consumption analytics.

C. Platform Administrator (Super Admin Portal)

- Global platform dashboard with cross-tenant metrics and active restaurant counts.
- Restaurant tenant governance: inspect, approve, or suspend restaurant accounts.
- Platform user management: create, manage, and toggle status of restaurant owners and staff.
- Database maintenance and platform audit logging.

D. Future Scope

- **AI-Powered Dietary Recommendations:** Machine learning recommendation models based on past ordering habits and target calorie budgets.

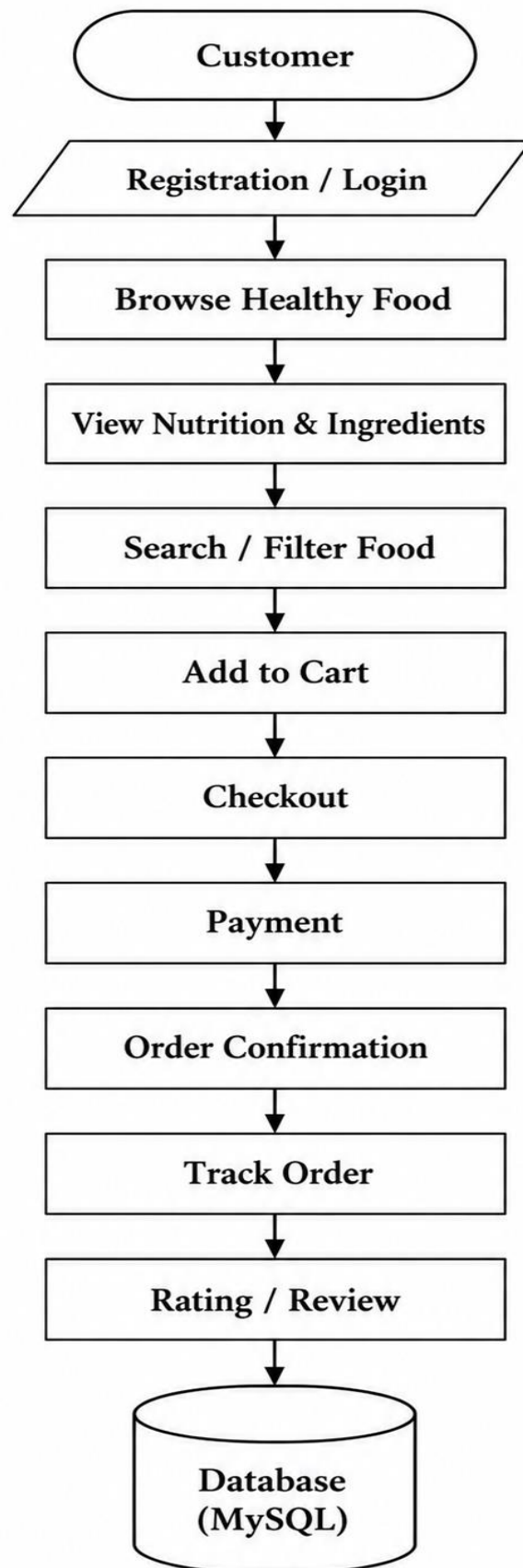
- **Personalized Health Profile Sync:** Integration with wearable fitness platforms (Apple Health, Google Fit) to log meal macros directly to health tracking apps.
- **Direct Banking Payment Gateway:** Integration with production payment aggregators (e.g., Razorpay, Stripe) for real-time online fund settlements.
- **Automated Stock Inventory & Recipe Deductions:** Ingredient-level warehouse tracking that automatically decrements raw ingredient quantities as orders are fulfilled.
- **Multilingual Menu Localization:** Support for regional Indian languages to expand customer accessibility.

5. Proposed Methodology

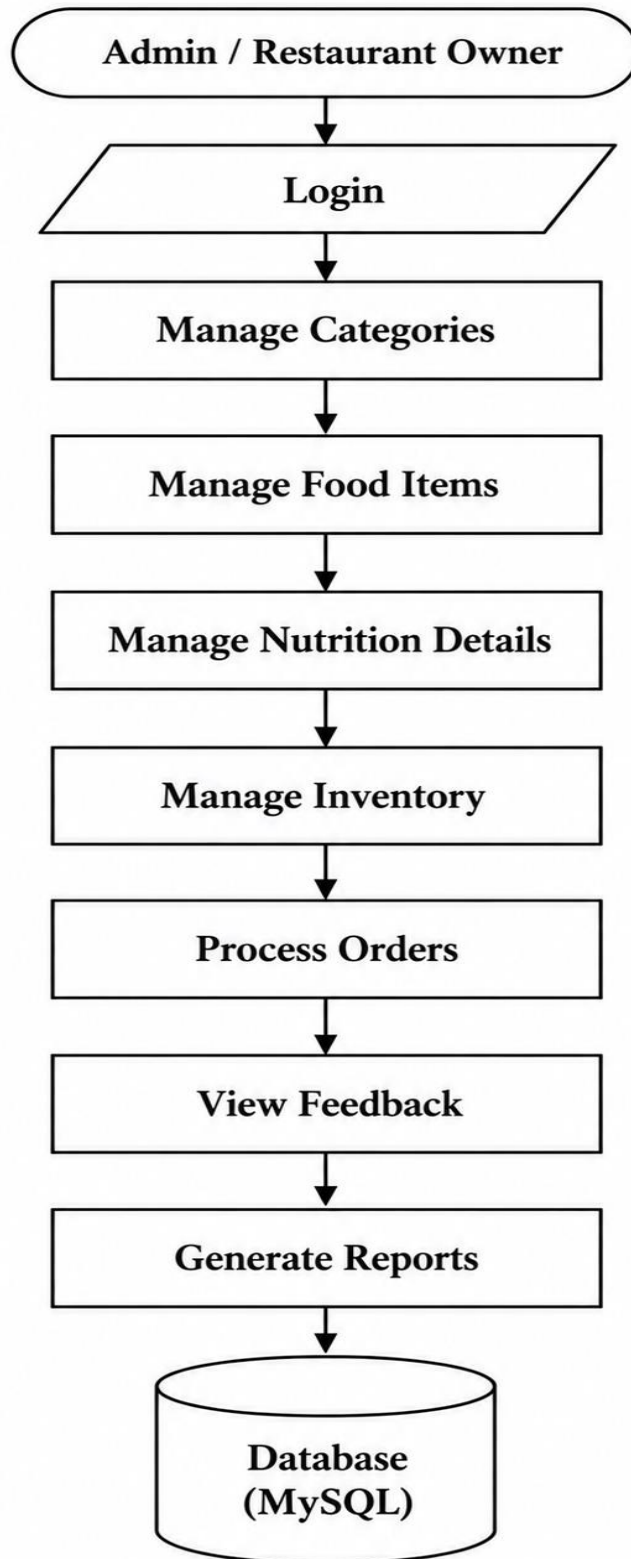
Healthy Bite is built following a custom 3-Tier Model-View-Controller (MVC) software architecture:

1. **Presentation Tier:** A responsive, mobile-first customer interface and desktop-optimized owner/admin dashboards built using HTML5, CSS3 custom variables, Vanilla JavaScript (ES6+ modular state pattern), and Bootstrap Icons.
2. **Application & Business Logic Tier:** Powered by PHP 8.0+ featuring a Front Controller pattern (`index.php`), custom REST routing (`routes/web.php` and `routes/api.php`), and dedicated Domain Service classes (`NutritionService`, `CartService`, `OrderService`, `PricingService`, `QrService`, `PaymentService`). Security is strictly enforced using CSRF tokens (`hash_equals`), bcrypt password hashing, and role-checking middleware.
3. **Data Tier:** A normalized MariaDB/MySQL relational database enforcing data integrity via InnoDB foreign key constraints, UTF-8 mb4 encoding, and 100% PDO prepared statements to eliminate SQL injection vulnerabilities.

System Workflow



Admin / Restaurant Workflow



6. SYSTEM MODULES

No.	Module	Main Functions
1	Customer Module	Registration, login, profile, browse food, search, filter
2	Food/Menu Module	Display food items, categories, ingredients and food details
3	Nutrition Module	Calories, protein, carbohydrates, fats, vitamins, minerals, allergens and dietary labels
4	Cart Module	Add food, update quantity, remove food and calculate cart total
5	Order Module	Checkout, place order, order confirmation and order tracking
6	Payment Module	Manage and record online payment information
7	Feedback & Review Module	Customer ratings and reviews
8	Admin/Restaurant Management Module	Manage users, categories, food, nutrition, inventory, orders, feedback and reports

7. DB / TABLES

1. admin :

Record Preview:

id	name	created at	updated at
1	Platform Super Admin	2026-08-30 15:50:38	2026-08-30 15:50:38

2. branches :

Record Preview:

id	restaurant id	name	address	city	state	phone	status
1	1	Greenhouse Kitchen - Indiranagar	100 Feet Road, HAL	Bengaluru	Karnataka	+918012345678	active

			2nd Stage, Indiranagar				
--	--	--	---------------------------	--	--	--	--

created at	updated at
2026-09-14 14:17:24	2026-09-14 14:17:24

3. categories :

Record Preview:

id	restaurant_id	name	slug	description	image	sort_order	status
1	1	Bowls	bowls	Nutrient-packed warm and cold signature bowls	cat_bowls.jpg	1	active

created at	updated at
2026-09-14 14:17:24	2026-09-14 14:17:24

4. customers :

Record Preview:

id	name	mobile	email	created at	updated at
1	Rahul Sharma	+919876543210	rahul.sharma@example.com	2026-09-14 14:17:24	2026-09-14 14:17:24

5. food_customizations :

Record Preview:

id	food_item_id	name	description	group_name	price_adjustment	calories_adjustment	protein_adjustment
11	13	Extra Protein Scoop / Serving	Additional protein portion	Portion Add-on	40.00	80	12.00

carbs_adjustment	fat_adjustment	fiber_adjustment	sugar_adjustment	sodium_adjustment	caffeine_adjustment	is_required	min_quantity
2.00	1.50	0.50	0.00	120.00	0.00	0	0

max_quantity	is_available	sort_order	created at	updated at
1	1	1	2026-09-14 14:17:24	2026-09-14 14:17:24

6. food_items :

Record Preview:

id	restaurant_id	category_id	name	slug	description	image	ingredients
1	1	1	Paneer Protein Bowl	paneer-protein-bowl	Tandoori paneer, brown rice, fresh veggies	paneer_protein_bowl.jpg	Paneer, brown rice, lettuce, cucumber, tomato, corn, mint dressing

allergens	food_type	base_price	calories	protein	carbs	fat	fiber
Milk	vegetarian	249.00	520	28.00	54.00	20.00	8.00

sugar	sodium	caffeine	serving_size	is_featured	is_popular	is_available	created_at
6.00	620.00	0.00	450 g	1	1	1	2026-09-14 14:17:24

updated_at
2026-09-14 14:17:24

7. food_variants :

Record Preview:

i_d	food_item_id	name	description	price_adjustment	calories_adjustment	protein_adjustment	carbs_adjustment
5	13	Standard Portion	Regular serving size	0.00	0	0.00	0.00

fat_adjustment	fiber_adjustment	sugar_adjustment	sodium_adjustment	caffeine_adjustment	is_required	is_available	sort_order
0.00	0.00	0.00	0.00	0.00	1	1	1

created_at	updated_at
2026-09-14 14:17:24	2026-09-14 14:17:24

8. order_item_customizations :

Record Preview:

i_d	order_item_id	customization_id	customization_name_snapshot	quantity	price_adjustment	calories_adjustment	protein_adjustment
1	1	1	Extra Paneer	1	40.00	80	6.00

carbs_adjustment	fat_adjustment	fiber_adjustment	sugar_adjustment	sodium_adjustment	caffeine_adjustment	created at
2.00	4.00	1.00	0.50	90.00	0.00	2026-09-14 14:17:24

9. order_items :

Record Preview:

i d	order_id	food_item_id	food_name_snapshot	base_price_snapshot	variant_name_snapshot	variant_price_snapshot	quantity
1	1	1	Paneer Protein Bowl	249.00	Large	50.00	2

unit_price	total_price	calories	protein	carbs	fat	fiber	sugar
339.00	678.00	630	36.00	64.00	24.00	10.00	6.50

sodium	caffeine	created at
710.00	0.00	2026-09-14 14:17:24

10. orders :

Record Preview:

id	order_number	restaurant_id	branch_id	table_id	customer_id	order_type	subtotal
1	HB-1A09F232DD9-412	1	1	4	2	dine_in	798.00

tax	service_charge	total_amount	payment_status	order_status	notes	created at	updated at
39.90	0.00	837.90	completed	complete	Less spicy, please	2026-09-14 14:17:24	2026-09-14 14:45:10

11. payments :

Record Preview:

id	order_id	payment method	transaction reference	amount	status	paid at	created at
1	1	upi	TXN-UPI-8CBFCF82D856	837.90	completed	2026-09-14 14:45:05	2026-09-14 14:45:05

12. qr_tokens :

Record Preview:

i d	restaurant_i d	branch_i d	table_i d	token	statu s	expires_a t	created_a t
1	1	1	4	hb_qr_token_indiranagar_tbl4_7f8a9b	active	2027-09-14 23:59:59	2026-09-14 14:17:24

13. restaurant_tables :

Record Preview:

id	restaurant_id	branch_id	table_number	status	created_at	updated_at
1	1	1	Table 1	available	2026-09-14 14:17:24	2026-09-14 14:17:24

14. restaurants:

Record Preview:

i d	owner_user_ id	name	slug	logo	cover_image	descripti on	phone
1	2	Greenhouse Kitchen	greenhouse-kitchen	greenhouse_logo.png	greenhouse_cover.jpg	Healthy casual dining with nutrition-aware meals	+918012345678

email	address	city	state	status	created_at	updated_at
hello@greenhousekitchen.com	100 Feet Road, HAL 2nd Stage, Indiranagar	Bengaluru	Karnataka	approved	2026-09-14 14:17:24	2026-09-14 14:17:24

15. reviews :

Record Preview:

id	restaurant_id	customer_id	order_id	rating	comment	restaurant_reply	replied_at
1	1	1	1	5	Fresh food, fantastic taste!	Thank you!	2026-09-14 18:10:00

status	created_at	updated_at
approved	2026-09-14 17:45:00	2026-09-14 18:10:00

16. roles :

Record Preview:

id	name	slug	description	created_at	updated_at
1	Super Admin	super_admin	Platform administrator with full privileges	2026-09-14 14:17:24	2026-09-14 14:17:24

17. users :

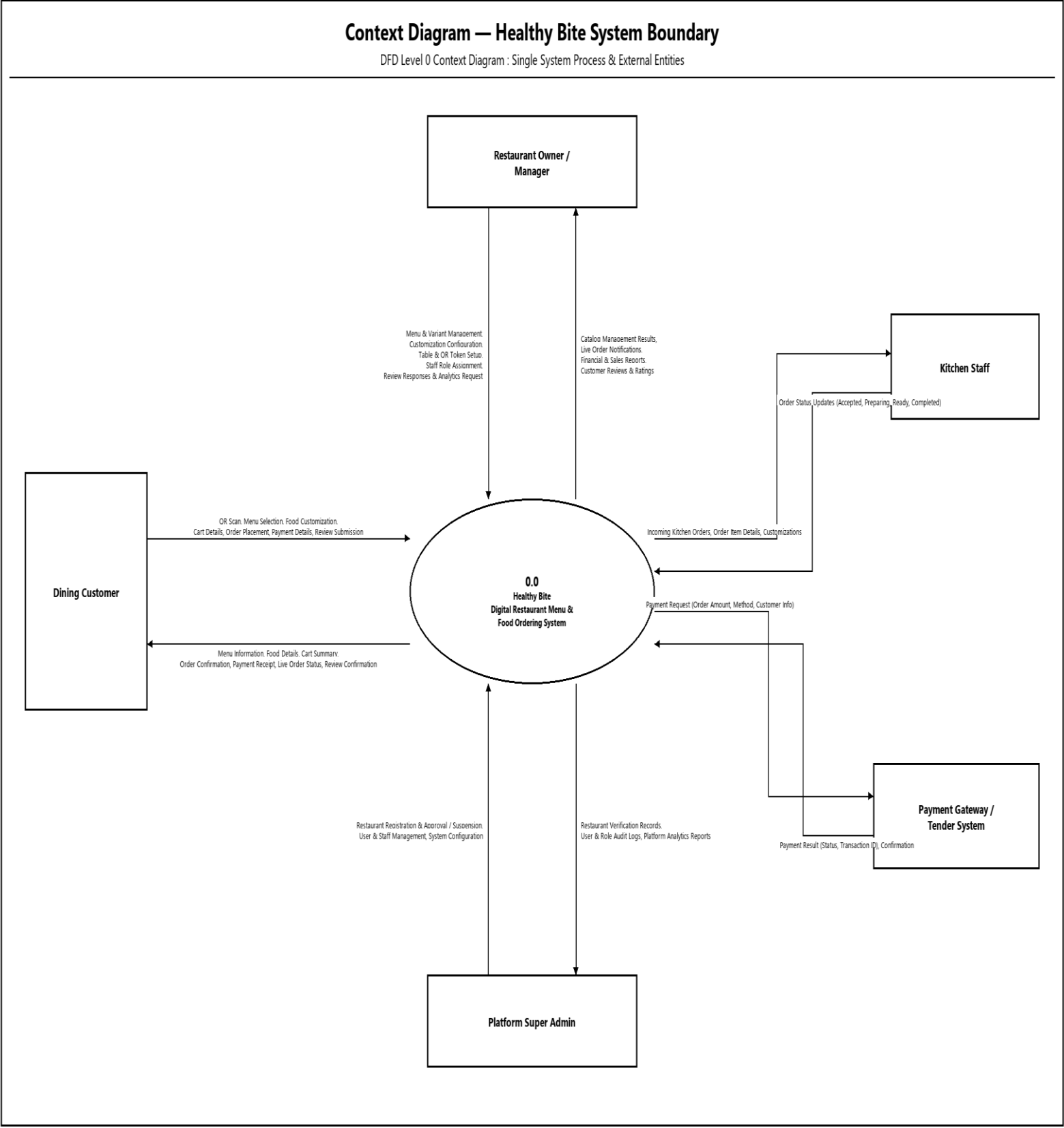
Record Preview:

id	role_id	restaurant_id	name	email	password	status	created_at
1	1	1	System Admin	admin@healthybite.com	\$2y\$12\$***** (bcrypt)	active	2026-09-14 14:17:24

updated_at
2026-09-14 14:17:24

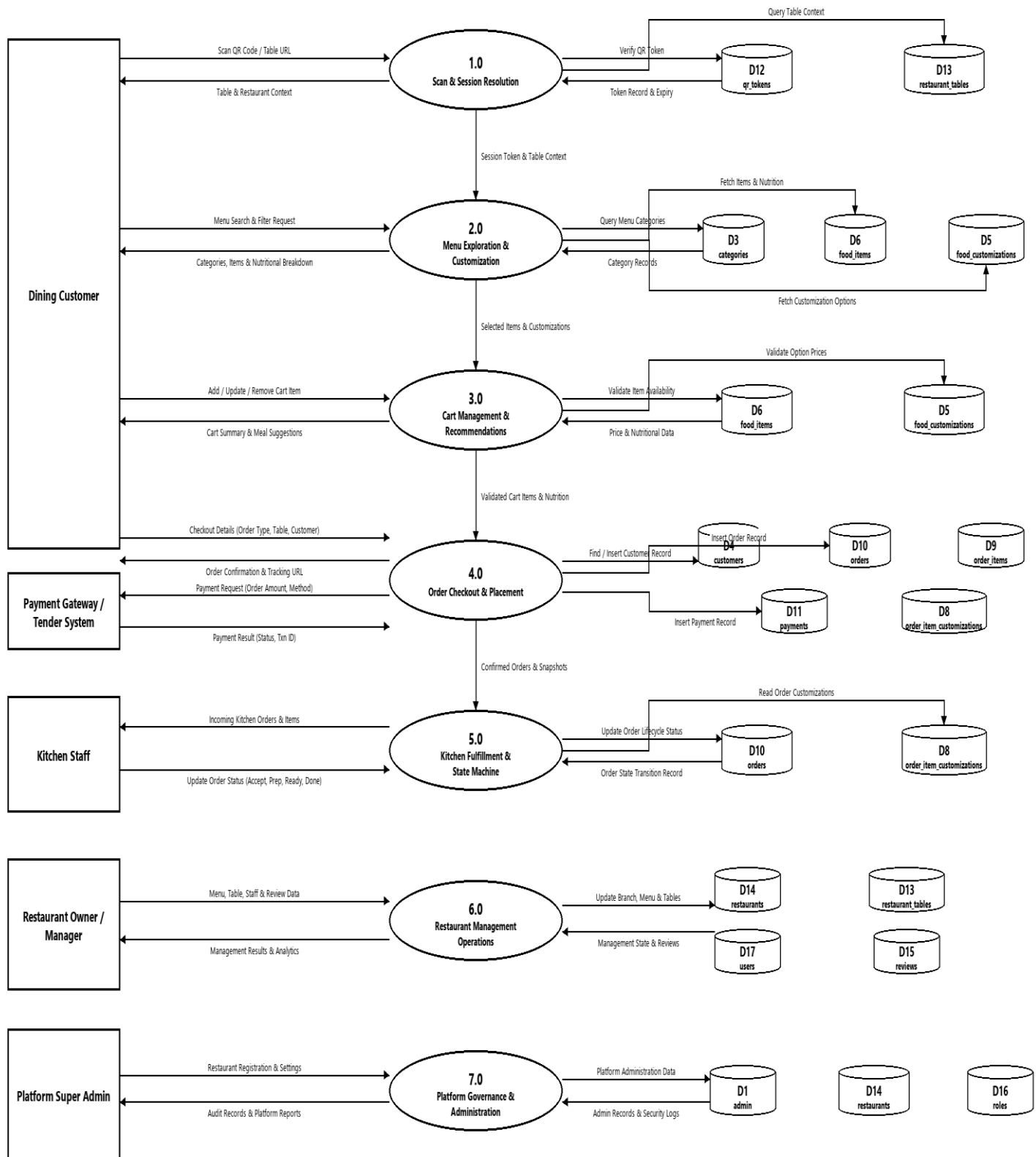
8. Diagrams

8.1 Data Flow Diagram (DFD)



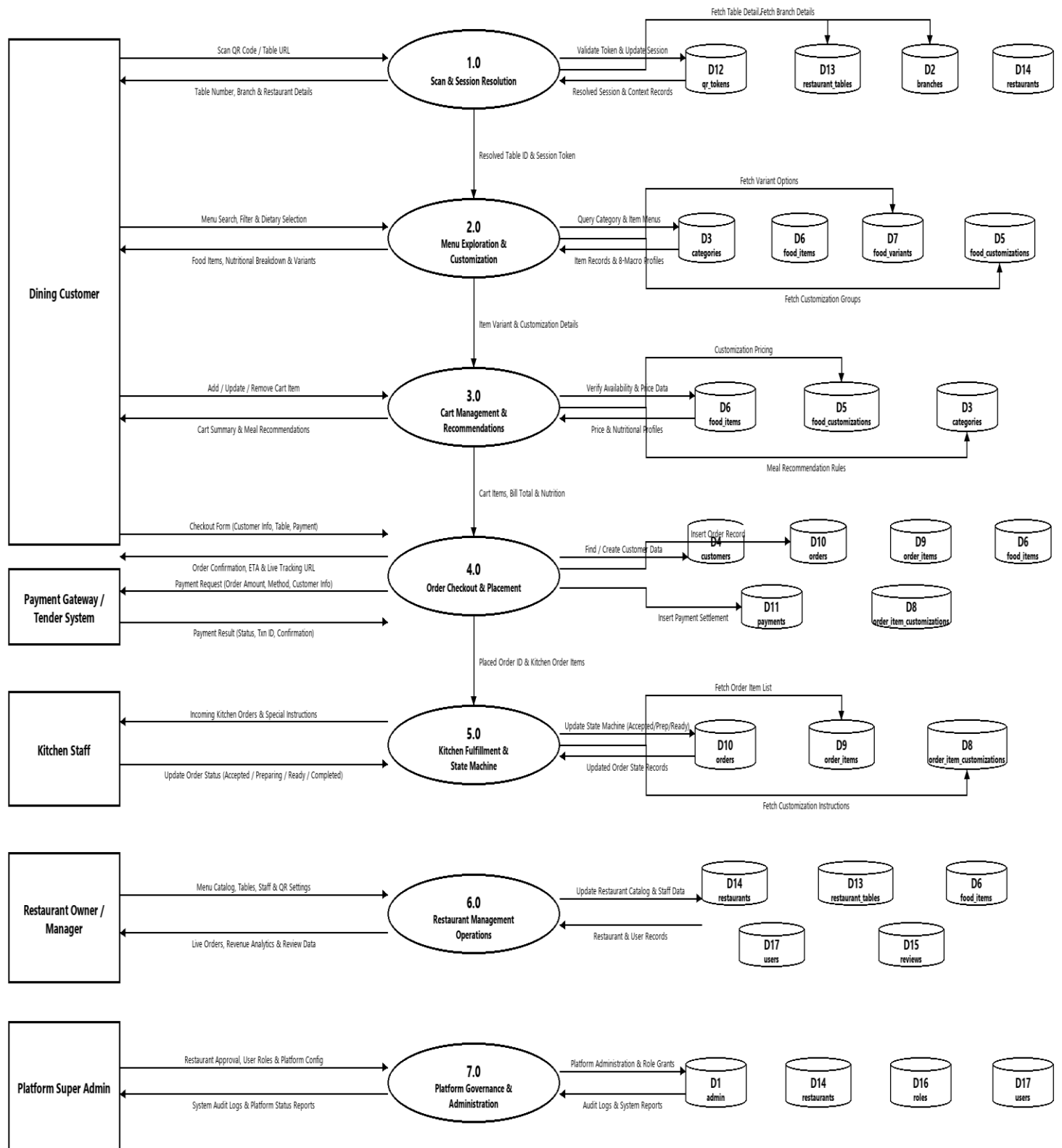
DFD Level 0 — System Functional Decomposition

Healthy Bite — High-Level Functional Modules & Entity Interfacing



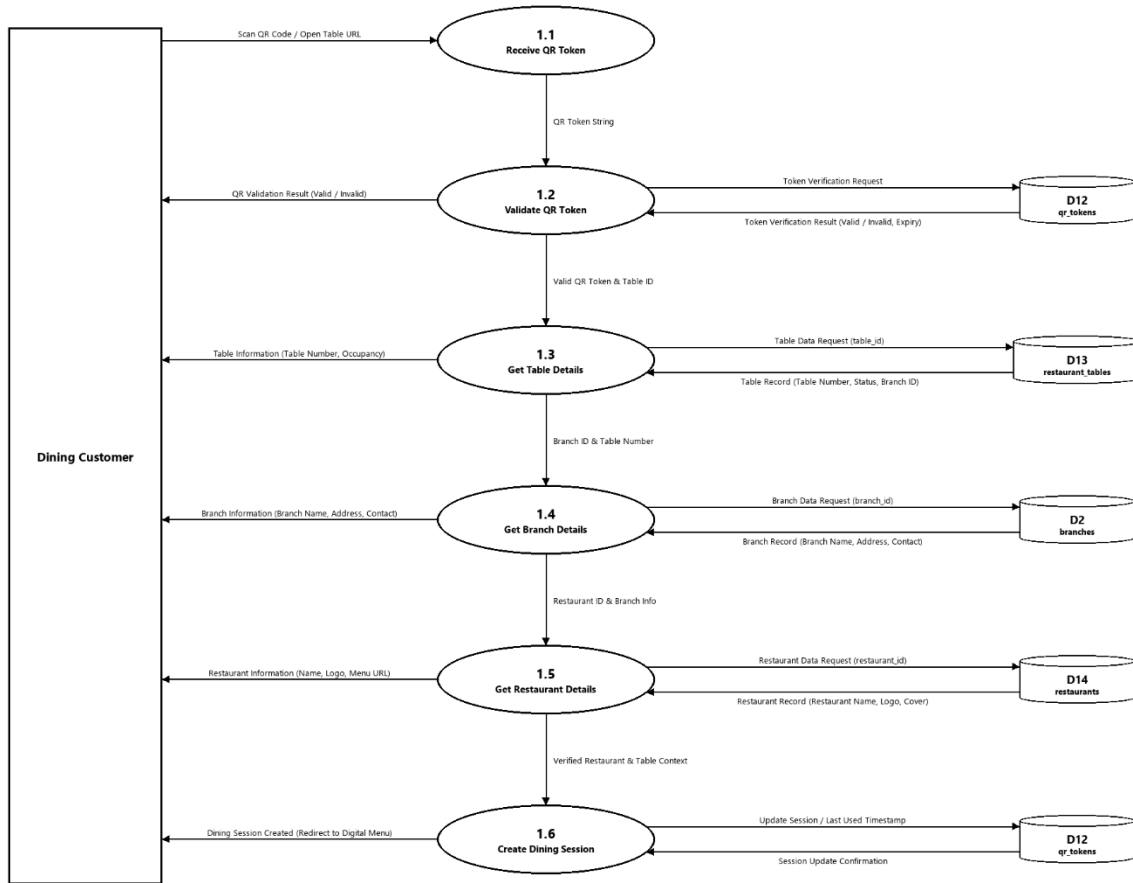
DFD Level 1 — Subsystem Operational Flow Diagram

Healthy Bite — Comprehensive Data Store Read/Write Balancing



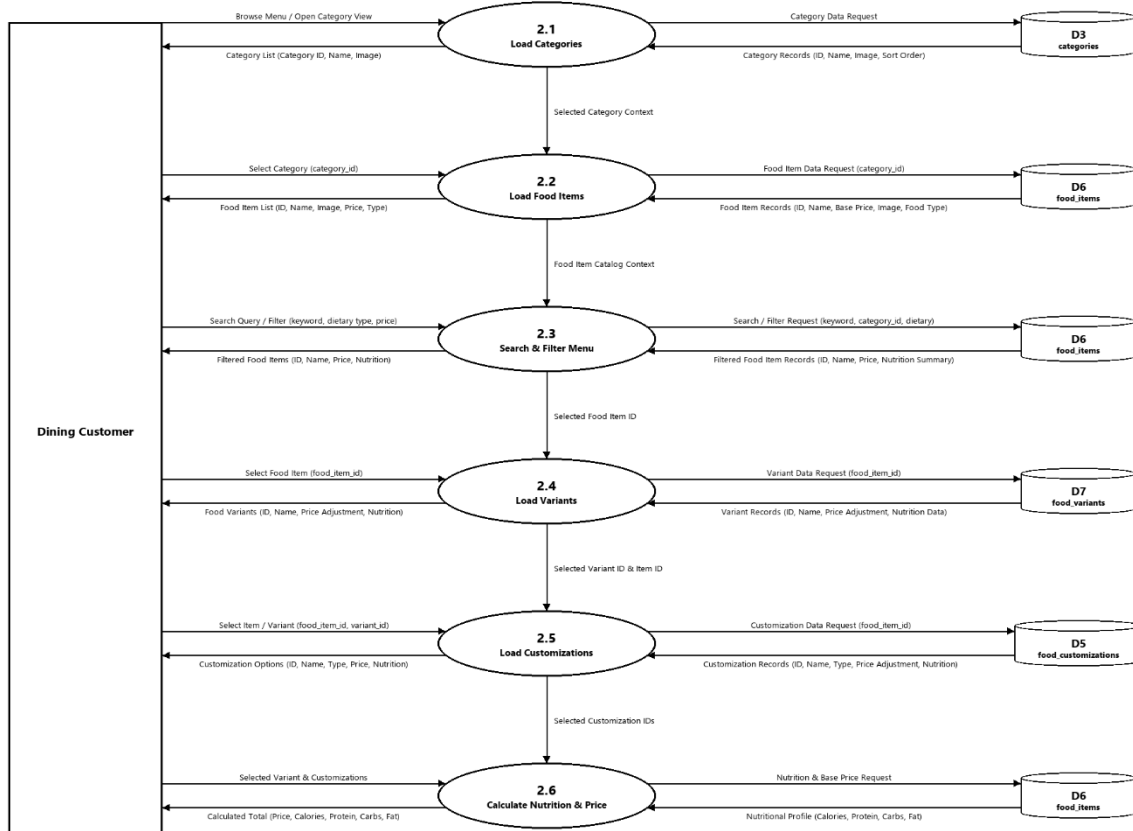
DFD - 2 : Level 2 (1.0 Scan & Session Resolution)

Subsystem Decomposition : Cryptographic QR Token Resolution & Dining Session Initialization



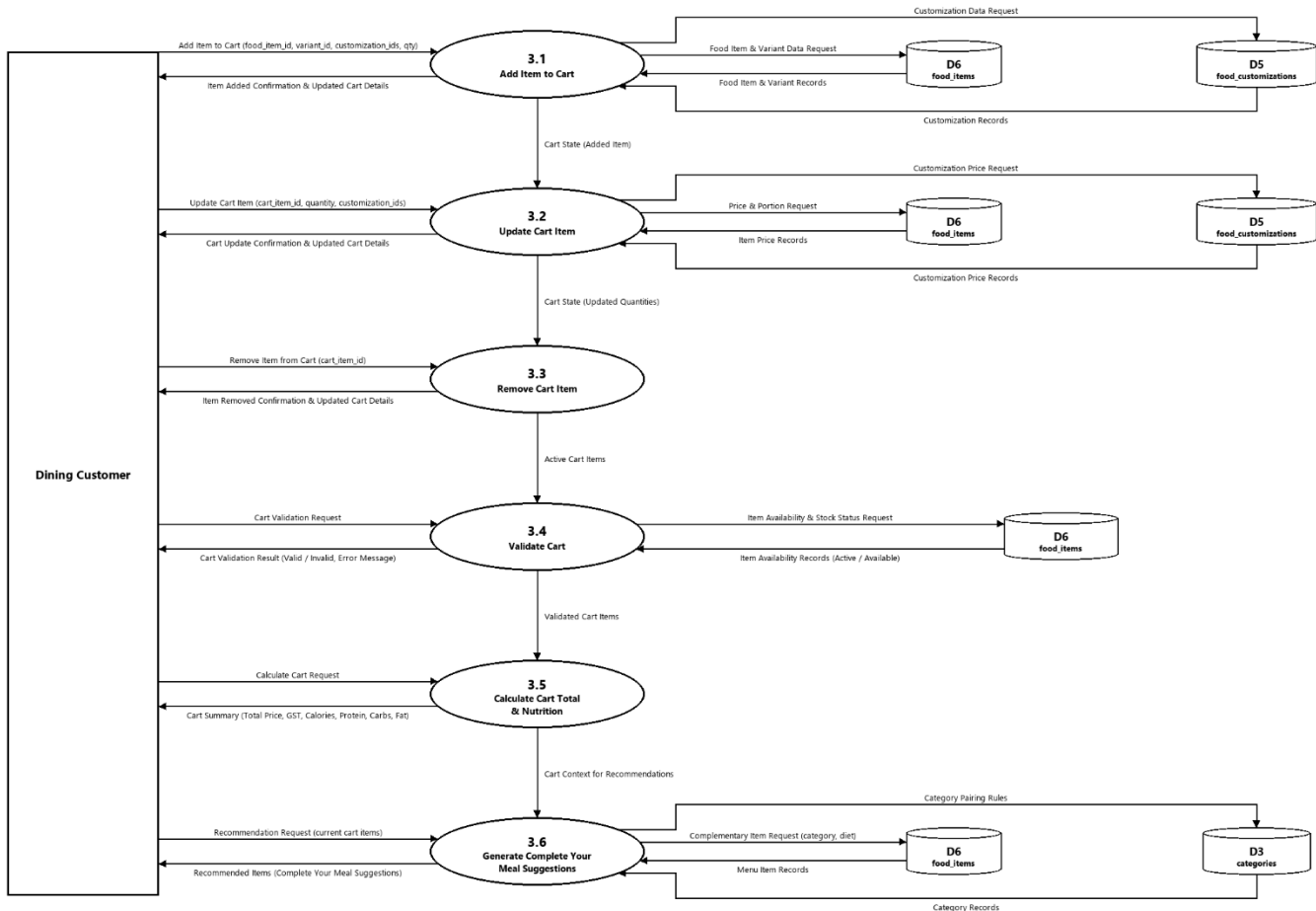
DFD - 3 : Level 2 (2.0 Menu Exploration & Customization)

Subsystem Decomposition : Menu Catalog, Multi-Variant Configuration & 8-Macro Engine



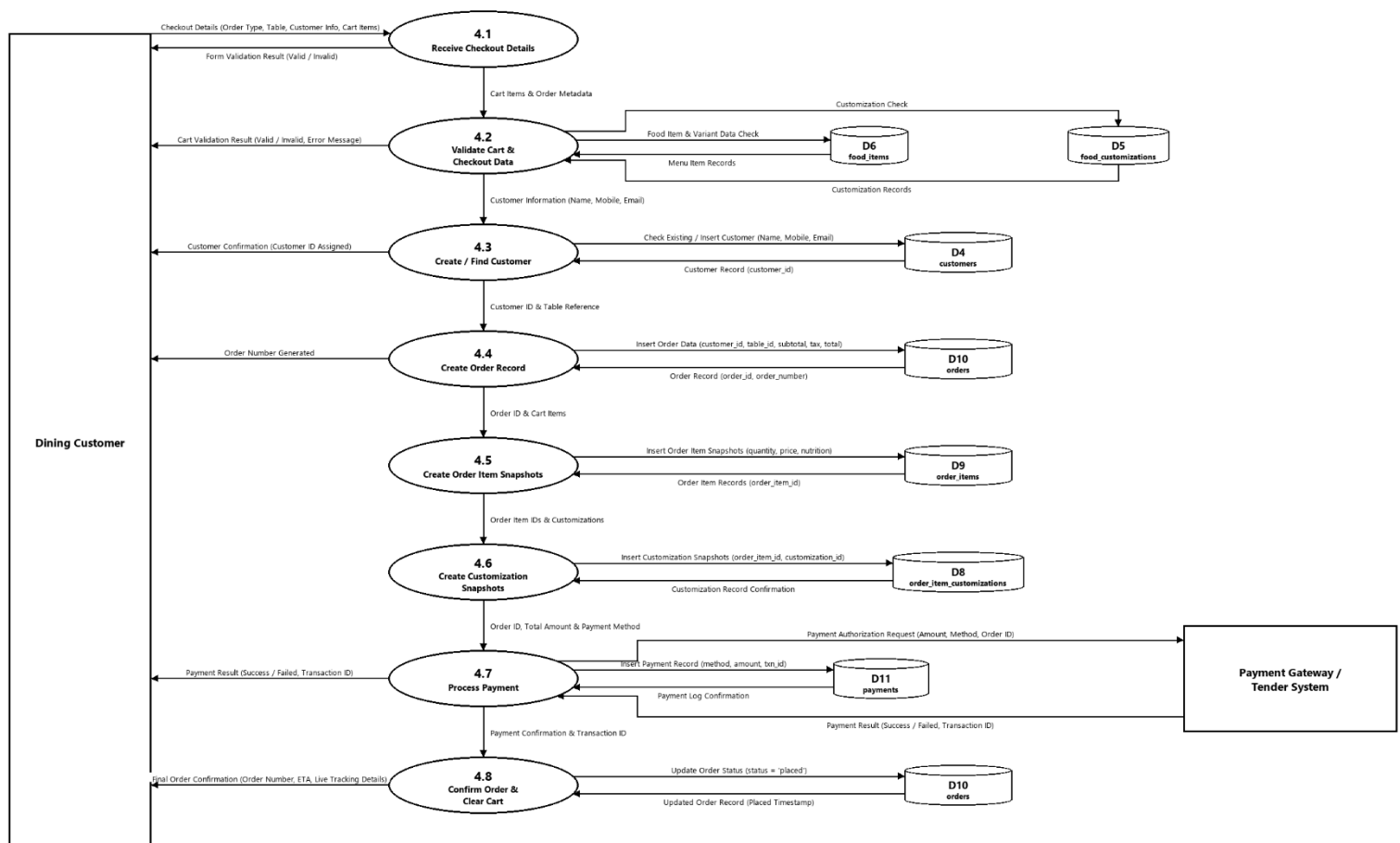
DFD - 4 : Level 2 (3.0 Cart Management & Recommendations)

Subsystem Decomposition : Client Cart State, Dynamic Totals & Recommendation Engine



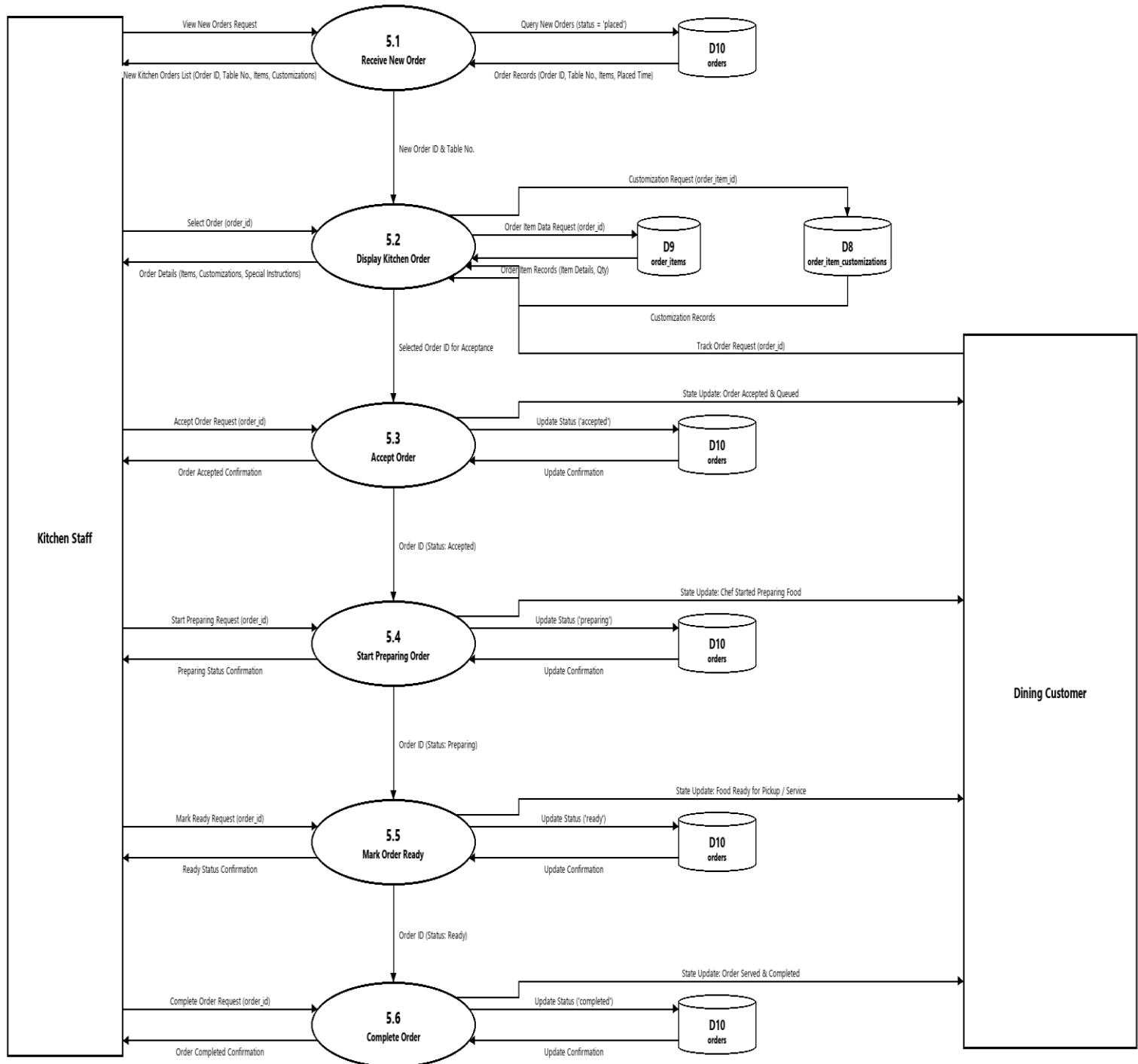
DFD - 5 : Level 2 (4.0 Order Checkout & Placement)

Subsystem Decomposition : Transactional Order Insertion, Snapshots & Payment Gateway Integration

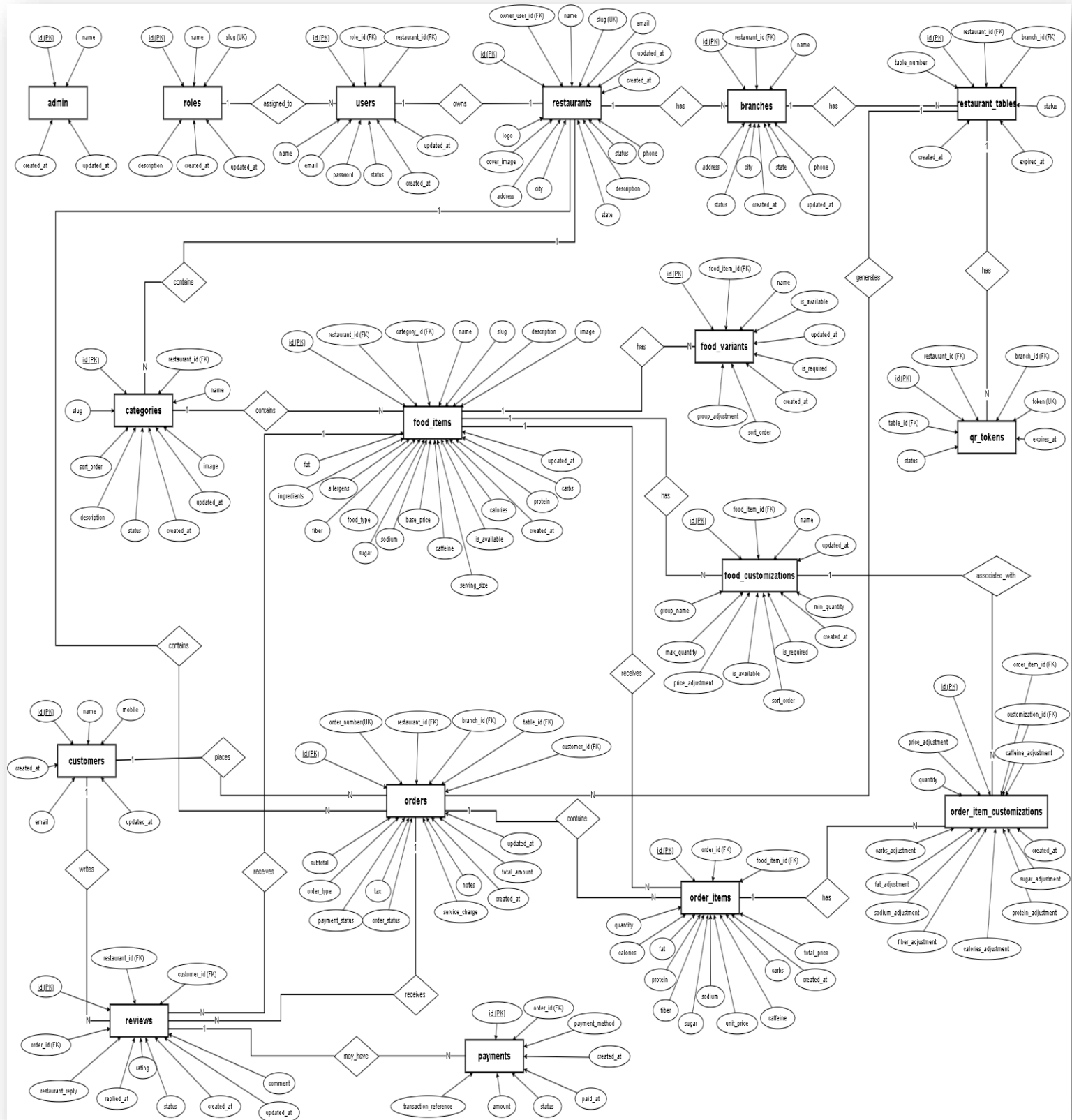


DFD - 6 : Level 2 (5.0 Kitchen Fulfillment & State Machine)

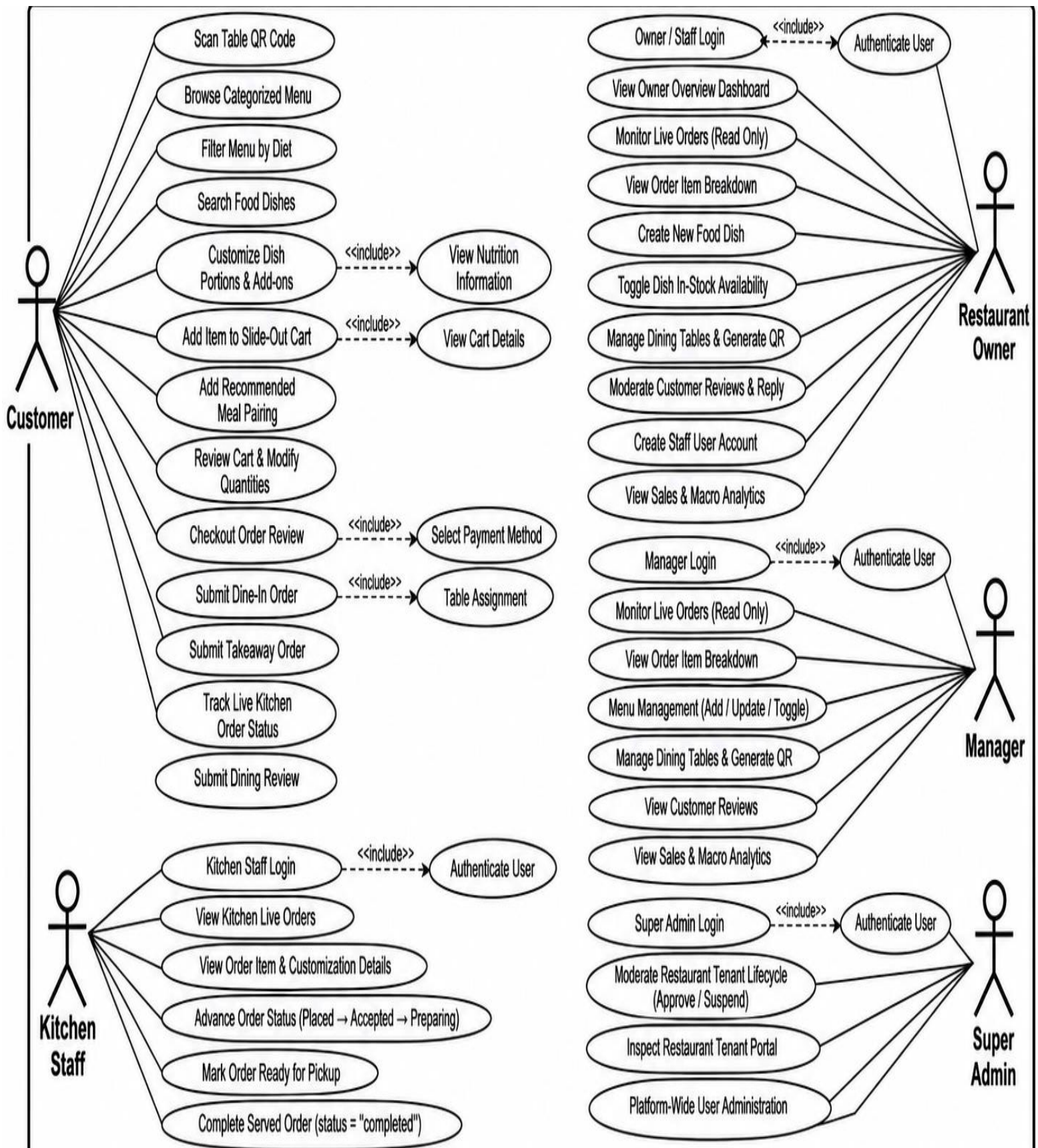
Subsystem Decomposition : Kitchen Kanban Lifecycle, State Transitions & Customer Notifications



8.2 Entity Relationship Diagram (ERD)



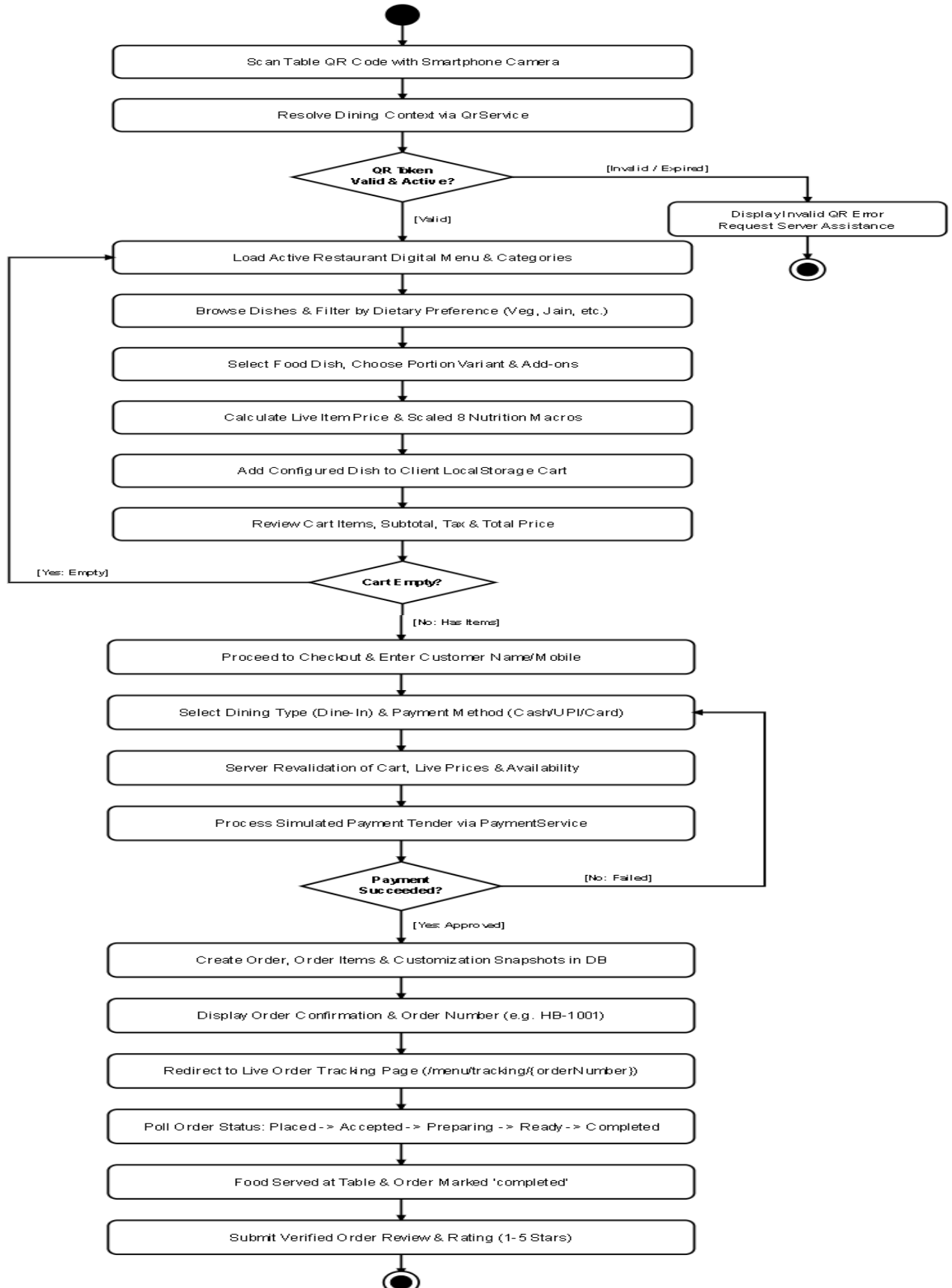
8.3 Use Case Diagram



8.4 Activity Diagram

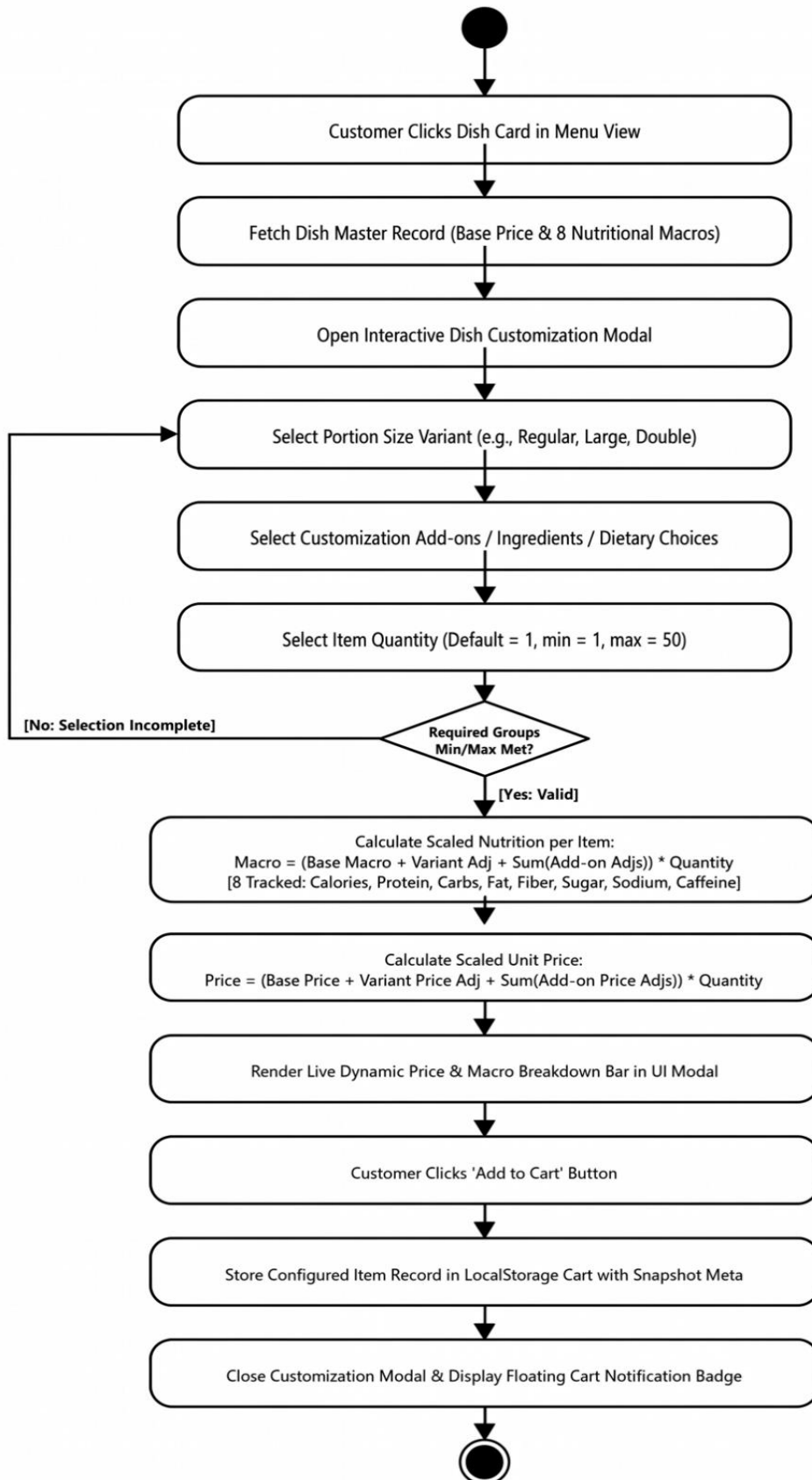
AD-01 : CUSTOMER ORDERING WORKFLOW

UML Activity Diagram • End-to-End Table QR Dining, Cart Validation, Tracking & Review



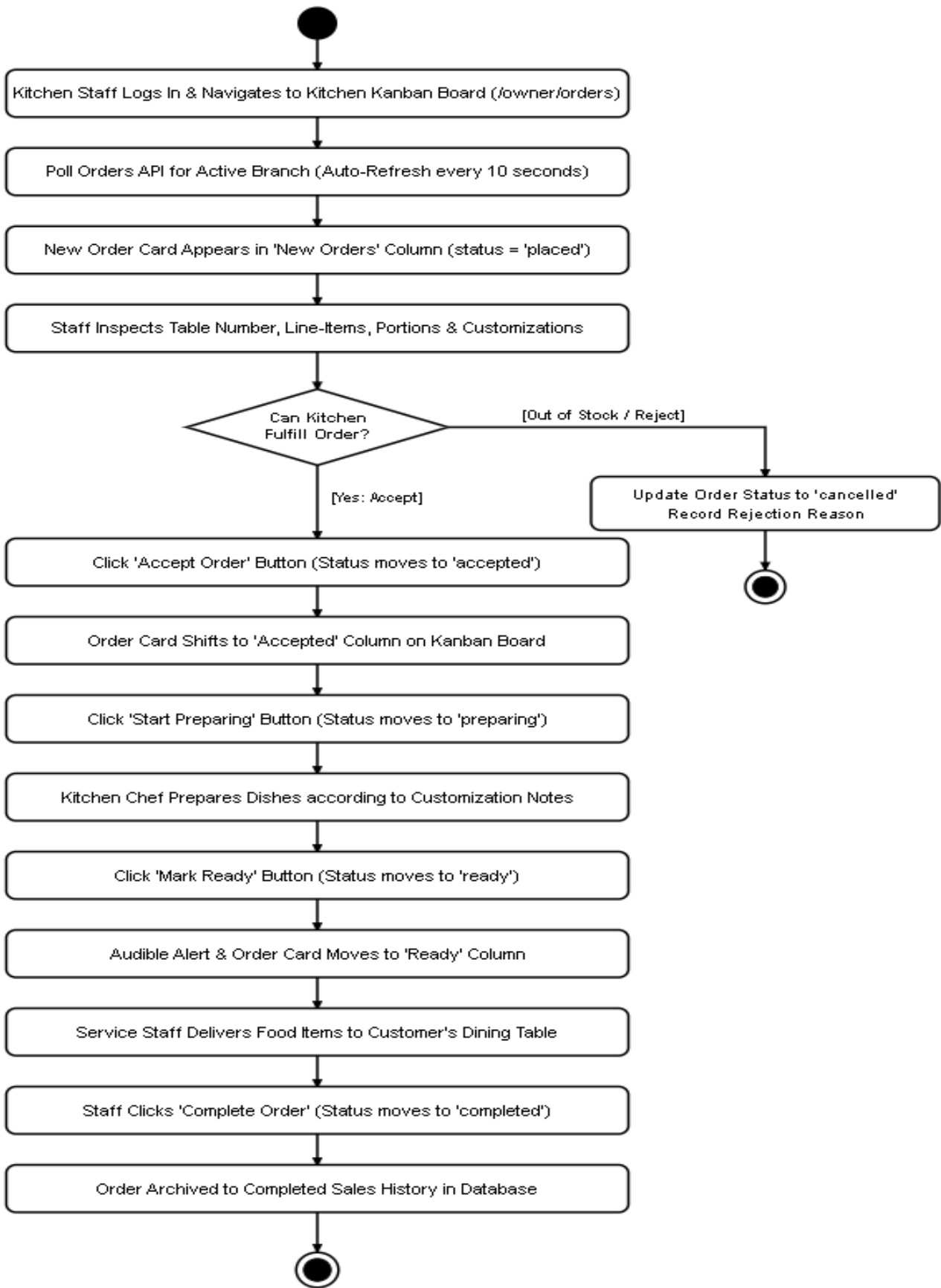
AD-02: FOOD CUSTOMIZATION & PRICE/NUTRITION CALCULATION

UML Activity Diagram • Variant Scaling, Customization Deltas & All 8 Nutritional Macros



AD-03: KITCHEN LIVE ORDER MANAGEMENT

UML Activity Diagram • Kitchen Kanban Transitions (placed -> accepted -> preparing -> ready -> completed)



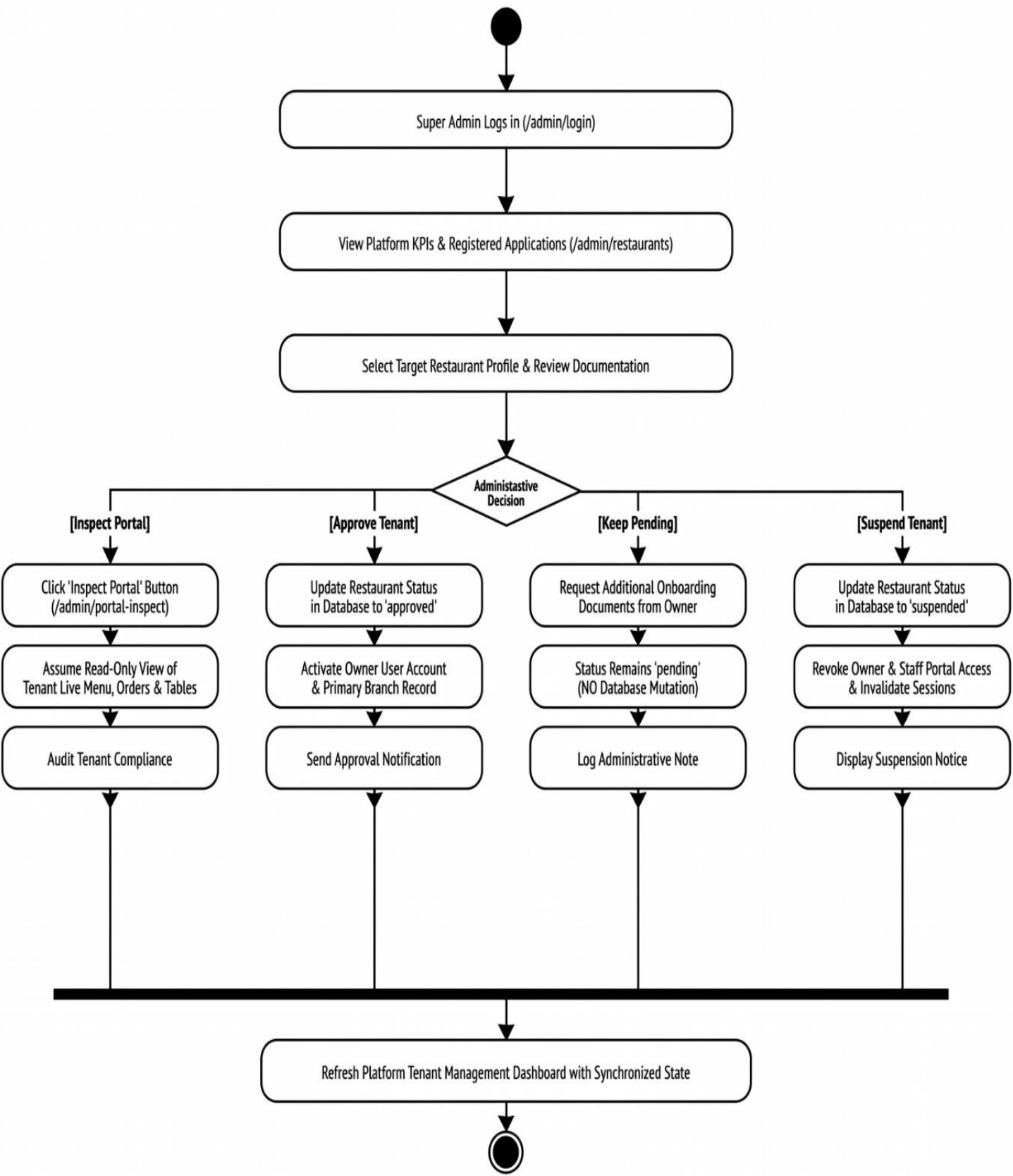
AD-04: RESTAURANT OWNER FOOD & MENU MANAGEMENT

UML Activity Diagram • Category CRUD, Dish Creation, 8 Macros, Portion Variants & Availability Toggles



AD-05: SUPER ADMIN RESTAURANT & PLATFORM MANAGEMENT

UML Activity Diagram • Multi-Tenant Onboarding, Approval, Portal Inspection & Suspension Lifecycle



9. TESTING

9.1 TEST PLAN

The test plan was prepared to verify the major functions and technical controls of the Healthy Bite digital restaurant menu and food ordering system. It covers the customer QR ordering journey, food catalog and customization, price and nutrition calculation, cart and checkout validation, order and payment processing, kitchen status management, live order tracking, restaurant owner functions, Super Admin functions, database integrity, authentication and authorization, security controls, responsive and browser-facing behavior, dynamic flows, and release-level validation.

The completed master QA cycle executed 74 test cases against the running PHP application, live MySQL database, application services, routes, and browser-facing functionality. The final result was 74/74 tests passed, with four defects identified, corrected, and retested before release validation.

9.2 TESTING APPROACH

The testing approach was derived from the completed Healthy Bite implementation and the final master QA testing report. The test cases were organized using the same academic testing structure used in the Internship Tracker research paper reference: each test case records a Test Case ID, Input / Test Action, Expected Output, Actual Output, and Status. The testing is divided into functional, integration, database, security, end-to-end, dynamic-flow, and release-validation groups so that each major project module is represented clearly.

Testing was performed in the following order: environment and static checks, database verification, unit/business-logic checks, API validation, authentication and role-based access control, customer end-to-end workflow, owner and kitchen workflow, Super Admin workflow, defensive security testing, dynamic flow testing, catalog/media checks, and final release/regression checks.

9.3 TESTING SUMMARY

Testing Area	Functionality Covered	Status
Environment & Static Analysis	PHP environment, PDO driver, configuration, PHP lint, JavaScript compilation	Successfully Executed
Database Integrity	17 tables, 208 columns, 26 foreign keys, payment mapping, orphan scan	Successfully Executed
Unit / Business Logic	Pricing, GST, 8-macro nutrition, caffeine, NULL preservation, QR and cart logic	Successfully Executed

Testing Area	Functionality Covered	Status
API Testing	Health, restaurant, categories, foods, cart validation, invalid resource handling	Successfully Executed
Authentication & RBAC	Owner/admin protection, invalid credentials, role verification, removed route	Successfully Executed
Customer End-to-End	Order creation, snapshots, simulated payment, status, tracking	Successfully Executed
Owner & Kitchen	Kanban transitions, tables, QR/table status, review replies, analytics	Successfully Executed
Security & Integrity	SQLi, XSS, CSRF, IDOR/tenant isolation, immutable snapshots, concurrency	Successfully Executed
Dynamic Flow / Catalog	Repository dynamic functions, table redirect, media URLs, dietary types	Successfully Executed
Release Validation	HEAD routing, order modal, menu CRUD, legacy feature exclusion	Successfully Executed

9.4 TESTING NOTES

- The test-case tables follow the five-column format: Test Case ID, Input / Test Action, Expected Output, Actual Output, and Status.
- The Actual Output column records the observed and verified behavior from the completed Healthy Bite Master QA validation report; it is not a placeholder result.
- The testing scope includes functional, integration, unit/business-logic, database, security, concurrency, dynamic-flow, browser/API, and release-validation checks.
- Payment is intentionally tested as simulated payment for supported methods. Third-party banking gateway integration is outside the implemented project scope.
- Live customer tracking is tested through the implemented HTTP polling mechanism.
- The legacy admin table is treated as a database artifact and is not the active administrator authentication mechanism.
- All 74 executed test cases passed after four defects were corrected and regression-tested.

9.5 TEST CASES

9.5.1 Customer / Core / API Test Cases

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-01	PHP 8.2+ Compatibility	PHP version is >= 8.2 (Running: PHP 8.5.10)	PHP 8.5.10 detected; compatibility requirement satisfied	Successfully Executed
T-02	PDO MySQL Driver	pdo_mysql extension active and loaded	pdo_mysql loaded successfully	Successfully Executed
T-03	Live Database Connection	PDO connection to healthy_bite responds SELECT 1 = 1	Live database connection responded successfully	Successfully Executed

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-04	.env Configuration	.env present, loaded and parseable via App\Core\Env	.env loaded and parsed successfully	Successfully Executed
T-05	Core Constants	ROLE_*, ORDER_*, TABLE_*, FOOD_* constants defined	Required core constants verified	Successfully Executed
T-06	PHP Syntax Validation	All 94 project PHP files pass php -l lint with zero errors	All 94 PHP files passed php -l	Successfully Executed
T-07	JavaScript VM Compilation	All 14 public JS files pass Node.js VM syntax compilation	All 14 JavaScript files compiled successfully	Successfully Executed
T-08	17 Physical Tables	All 17 documented tables verified in information_schema	All 17 physical tables verified	Successfully Executed
T-09	208 Database Columns	Exact column count across all 17 tables matches 208	208 columns verified across 17 tables	Successfully Executed
T-10	26 Foreign Keys	All 26 foreign key constraints verified active	All 26 foreign keys active and verified	Successfully Executed
T-11	Order-Payment Mapping 1:1	payments.order_id UNIQUE key strictly enforced (Non_unique=0)	Unique order-to-payment mapping verified	Successfully Executed
T-12	Orphan Record Scan	Zero orphan rows across all 26 FK relations	0 orphan rows found	Successfully Executed
T-13	Base + Variant + Custom Price	Base (100) + Var (20) + Custom (25) = Rs 145.00	Rs 145.00 calculated exactly	Successfully Executed
T-14	Line Total Scaling	Rs 145.00 x 3 = Rs 435.00 exact line total	Rs 435.00 line total calculated exactly	Successfully Executed
T-15	Order Totals & 5% GST	Subtotal Rs 435 + Tax Rs 21.75 = Rs 456.75	Rs 456.75 final total verified	Successfully Executed
T-16	8 Macros & Caffeine Sum	All 8 macros summed: Cal, Prot, Carb, Fat, Fib, Sug, Sod, Caff	All 8 nutrition values summed correctly	Successfully Executed
T-17	Caffeine in mg	Caffeine calculated in mg and never coerced to grams	Caffeine remained in mg throughout calculation	Successfully Executed
T-18	Quantity Scaling Nutrition	Nutrition doubles accurately when quantity = 2	Nutrition doubled accurately for quantity 2	Successfully Executed
T-19	Strict Preservation NULL	Unmeasured fields retain NULL, not coerced to 0	NULL nutrition fields remained NULL	Successfully Executed
T-20	Valid QR Token Resolution	Active token resolves restaurant, branch, and table IDs	Active QR token resolved correct context	Successfully Executed
T-21	Invalid QR Token Rejection	Random/expired token returns null context	Invalid or expired token returned null context	Successfully Executed
T-22	Empty Cart Rejection	Empty cart payload throws InvalidArgumentException	Empty cart rejected correctly	Successfully Executed
T-23	Cross-Restaurant Isolation	Food from Restaurant A rejected under Restaurant B context	Cross-restaurant food access blocked	Successfully Executed
T-24	GET /api/health	Returns HTTP 200 with status=ok and database=connected	HTTP 200 and healthy DB status returned	Successfully Executed
T-25	GET /api/restaurant	Returns HTTP 200 with active restaurant profile	Active restaurant profile returned	Successfully Executed

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-26	GET /api/categories	Returns HTTP 200 with list of active menu categories	Active category list returned	Successfully Executed
T-27	GET /api/foods	Returns HTTP 200 with complete menu items catalog	Complete menu catalog returned	Successfully Executed
T-28	GET /api/foods/1	Returns HTTP 200 with Paneer Bowl details & variants	Paneer Bowl details and variants returned	Successfully Executed
T-29	GET /api/foods/99999	Returns HTTP 404 for non-existent food item ID	HTTP 404 returned correctly	Successfully Executed
T-30	POST /api/cart/validate	Returns HTTP 200 for valid items & required groups	Valid cart returned HTTP 200 with verified totals	Successfully Executed
T-31	Cart Server Recalculation	Server ignores client fake prices; computes from DB	Fake prices ignored; authoritative DB values used	Successfully Executed
T-32	Owner Dashboard Protection	Unauthenticated /owner/dashboard redirects to /owner/login	Unauthenticated access redirected to owner login	Successfully Executed
T-33	Admin Dashboard Protection	Unauthenticated /admin/dashboard redirects to /admin/login	Unauthenticated access redirected to admin login	Successfully Executed
T-34	Invalid Credentials Check	Bad email/password fails authentication gracefully	Invalid credentials rejected cleanly	Successfully Executed
T-35	Removed Admin Orders Route	/admin/orders returns HTTP 404 (strictly eliminated)	HTTP 404 confirmed; route absent	Successfully Executed
T-36	Order Creation Pipeline	Order placed with valid customer, table, and branch	Order created successfully with valid context	Successfully Executed
T-37	Order Item Snapshot Freeze	food_name, base_price, unit_price saved in snapshot	Historical item snapshot persisted correctly	Successfully Executed
T-38	Customization Snapshot Freeze	customization_name and price_adjustment saved in snapshot	Customization snapshot persisted correctly	Successfully Executed
T-39	Payment Simulation (UPI)	Simulates payment, generates TXN ref, marks completed	UPI simulation completed with transaction reference	Successfully Executed
T-40	Status Advance on Payment	Order status advances from placed to accepted upon payment	Order advanced to accepted after payment	Successfully Executed
T-41	Order Tracking Polling	Live tracking finds order by order_number with status	Live tracking returned current order status	Successfully Executed

9.5.2 Owner / Kitchen / Admin Test Cases

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-42	Kitchen Kanban State Machine	placed->accepted->preparing->ready->completed	All required status transitions succeeded	Successfully Executed
T-43	Dining Tables 1-12 Active	All 12 dining tables present and queryable for Branch 1	Tables 1-12 verified and queryable	Successfully Executed
T-44	Table Status Toggle	Table toggles available <-> occupied seamlessly	Table status toggled successfully	Successfully Executed

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-45	Review Reply Persistence	Owner reply persisted to reviews table with replied_at	Reply persisted with replied_at timestamp	Successfully Executed
T-46	Super Admin Role Verification	Super Admin account verified with role_id = 1	Super Admin role verified	Successfully Executed
T-47	Restaurant Owner Role Verification	Owner account verified with role_id = 2	Restaurant Owner role verified	Successfully Executed
T-48	Kitchen Staff Role Verification	Kitchen Staff account verified with role_id = 4	Kitchen Staff role verified	Successfully Executed
T-49	Approved Restaurants Query	Platform lists approved active tenant restaurants	Approved tenant restaurants listed correctly	Successfully Executed

9.5.3 Security / Integrity / Concurrency Test Cases

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-50	SQL Injection: Search Filter	Malicious SQL in search handled via PDO prepared statements	Malicious SQL handled safely	Successfully Executed
T-51	SQL Injection: Query Params	IDs and filters parameterized; zero SQL injection vectors	IDs and filters safely parameterized	Successfully Executed
T-52	XSS Output Escaping	HTML/JS tags escaped via htmlspecialchars in views	Output escaping verified	Successfully Executed
T-53	CSRF Protection	State-changing POST actions protected by CSRF tokens	CSRF validation blocked unauthorized state changes	Successfully Executed
T-54	IDOR Tenant Protection	Cross-restaurant cart and order modification blocked	Cross-tenant modification blocked	Successfully Executed
T-55	Catalog Price Independence	Updating food base_price does not alter historical order snapshot	Historical order snapshot remained unchanged	Successfully Executed
T-56	Duplicate Payment Guard	Repeated payment request on paid order blocked gracefully	Repeated payment rejected cleanly	Successfully Executed

9.5.4 Dynamic Flow / Catalog / Release Test Cases

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-57	Find Table with Active Token	BranchRepository::findTableByNumber locates table + token	Table and token located correctly	Successfully Executed
T-58	Find First Available Table	BranchRepository::getFirstAvailableTable returns active table	Active table returned correctly	Successfully Executed
T-59	Table Number Redirect	HTTP GET /table/12 redirects (302) to /menu?token=...	HTTP 302 redirect verified	Successfully Executed
T-60	Invalid Token Error Banner	HTTP GET /menu with invalid token displays error notice	Error notice displayed correctly	Successfully Executed

Test Case ID	Input / Test Action	Expected Output	Actual Output	Status
T-61	Category Auto-Slug Generation	CategoryRepository::createCategory generates slug automatically	Slug generated automatically	Successfully Executed
T-62	Category Deletion Guard	Category deletion blocked if active food items are attached	Deletion guard blocked invalid removal	Successfully Executed
T-63	Staff Record Update	StaffRepository::updateStaff dynamically updates member details	Staff details updated successfully	Successfully Executed
T-64	Macro Sugar Aggregations	OrderRepository::getMacroStats computes total & avg sugar	Sugar totals and averages computed correctly	Successfully Executed
T-65	Checkout Table Dropdown	Checkout view renders verified active branch tables	Active branch tables rendered correctly	Successfully Executed
T-66	Owner Analytics Data Load	Analytics controller loads real AOV, revenue, orders	Real analytics data loaded successfully	Successfully Executed
T-67	Sugar Snapshot Integrity	OrderService creates order with accurate sugar snapshot	Sugar snapshot verified accurate	Successfully Executed
T-68	45 Unique Food Dish URLs	All 45 Unsplash dish image URLs return HTTP 200 OK	All 45 URLs returned HTTP 200	Successfully Executed
T-69	Distinct Photo Identifiers	All 45 food items use 100% distinct photo IDs	All 45 food items use distinct photo IDs	Successfully Executed
T-70	Food Dietary Types Valid	food_type enum values strictly vegetarian, vegan, etc.	Valid food_type enum values confirmed	Successfully Executed
T-71	HTTP HEAD Method Route Matching	Router maps HEAD to GET; curl -I returns HTTP 200 OK	HEAD request returned HTTP 200	Successfully Executed
T-72	Order Details Modal Retrieval	Order modal API retrieves items and customizations	Order details and customizations retrieved	Successfully Executed
T-73	Menu Item Create/Edit/Delete	Full food item CRUD lifecycle tested and restored	Full CRUD lifecycle passed	Successfully Executed
T-74	Legacy Features Clean Exclusion	No Call Servant or Platform Orders code exists	Removed legacy features remained absent	Successfully Executed

9.5.5 DEFENSIVE SECURITY TESTING RESULTS

Attack Vector / Scope	Security Test Applied	Protection Mechanism	Result
SQL Injection (SQLi)	Injected malicious SQL into search, query parameters and IDs.	PDO prepared statements with parameterized placeholders across repositories.	PASS
Cross-Site Scripting (XSS)	Injected script-like payloads into names, notes and customer review comments.	htmlspecialchars output escaping with ENT_QUOTES and UTF-8 in views.	PASS
Cross-Site Request Forgery	Dispatched state-changing POST requests without valid CSRF session tokens.	Csrf::validate() token verification rejects unauthorized state changes.	PASS
IDOR & Multi-Tenant Isolation	Attempted to order Food from Restaurant 1 under Restaurant 2 context via API payload.	CartService and SQL queries enforce strict restaurant_id multi-tenant scoping.	PASS

Attack Vector / Scope	Security Test Applied	Protection Mechanism	Result
Session & Auth Security	Attempted direct protected URL access to owner/admin portals without active session.	RestaurantMiddleware and AdminMiddleware redirect unauthenticated sessions to login.	PASS

9.5.6 VERIFIED DATABASE SCHEMA ARCHITECTURE

Table Name	Rows	Cols	FKs	Primary Key	Role / Constraints
admin	4	4	0	id	Legacy static administration table
branches	6	10	1	id	Restaurant dining locations
categories	16	10	1	id	Menu category hierarchy
customers	37	6	0	id	Registered and guest patrons
food_customizations	564	21	1	id	Dish modifiers and adjustments
food_items	47	25	2	id	Main culinary catalog and macros
food_variants	94	18	1	id	Portion sizes and variant adjustments
order_item_customizations	66	15	2	id	Frozen customization snapshot
order_items	43	19	2	id	Frozen item price/macro snapshot
orders	34	16	4	id	Core order transactions and statuses
payments	21	8	1	id	Tender settlement (UNIQUE: order_id)
qr_tokens	12	8	3	id	Table dining session tokens
restaurant_tables	12	7	2	id	Physical dining tables (1-12)
restaurants	3	15	1	id	Tenant restaurant accounts
reviews	7	11	3	id	Customer ratings and owner replies
roles	4	6	0	id	RBAC roles (admin, owner, manager, staff)
users	8	9	2	id	Authenticated employee and admin logins

9.6 FINAL TESTING STATEMENT

The Healthy Bite system was tested across its major customer, owner, kitchen, administrator, database, API, security, and release-validation functions. A total of 74 test cases were executed. All 74 test cases passed successfully after four identified defects were corrected and regression-tested. The final testing cycle found no unresolved critical defects, and the application was verified for final academic submission and release within the implemented project scope.

Test Execution	74 / 74 Passed (100%)
Defects	4 Identified, Fixed and Retested
Regression	All related suites passed after fixes
Final Status	VERIFIED & APPROVED

10. RESULTS

Results Summary Table

Result Area	Functional Evaluated	Scope	Observed Empirical Outcome	Validation Status
Customer Onboarding	Table QR resolution, fallback routing		Session created instantly; invalid tokens gracefully route to default menu	100% Passed
Menu & Dietary	Categories, dietary filters, search		Dynamic filtering by Veg/Vegan/Non-Veg/Jain; sub-100ms keyword search	100% Passed
Nutritional Transparency	8-macro display, allergen disclosure		Accurate calories, protein, carbs, fat, fiber, sugar, sodium, caffeine display	100% Passed
Cart & Customization	Variant selection, ingredient delta, edit		Portion scaling and in-cart customization editing recalculate price/macros	100% Passed
Order & Payment	Unique order generation, 5% GST, COD/simulated		Alphanumeric IDs created; mathematical tax computation verified	100% Passed
Live Kitchen Tracking	5-stage pipeline, background polling		State changes propagate to diner view within 4s without full-page reloads	100% Passed
Kitchen Operations	Kanban board, rapid order status transitions		Single-click state advance (Placed -> Accepted -> Preparing -> Ready)	100% Passed
Owner Administration	Menu CRUD, generation, analytics	slug	Automated slugging, media format validation, real-time KPI data loading	100% Passed
Database Integrity	17 tables, 208 columns, 26 foreign keys		Zero orphan records detected; historical order snapshots preserved	100% Passed
Defensive Security	SQLi, XSS, CSRF, IDOR, RBAC		Parameterized queries, output encoding, CSRF tokens, strict tenant isolation	100% Passed

11. CONCLUSION

11.1 Research and Implementation Synthesis

Modern online food ordering systems have achieved widespread commercial adoption by maximizing ordering convenience and delivery velocity. However, as established in the problem statement, this convenience has historically come at the expense of nutritional transparency. Mainstream platforms prioritize promotional discounts and rapid dispatch while treating food as a black-box commodity, offering negligible disclosure of calories, macronutrients, allergens, or ingredient origins. This information asymmetry directly undermines consumer health awareness and presents severe challenges for health-conscious diners, fitness enthusiasts, and individuals managing dietary conditions such as diabetes, obesity, and hypertension.

Healthy Bite: Digital Menu and Food Ordering System was conceived, designed, and developed to bridge this critical gap between digital food ordering convenience and proactive health management. The research and engineering effort demonstrates that full nutritional transparency can be seamlessly woven into every stage of the restaurant customer journey—from contactless QR table onboarding and ingredient customization to live kitchen tracking—without introducing operational friction for restaurant personnel or computational overhead for consumers.

11.2 Direct Alignment with Project Objectives

The completed implementation directly satisfies every core objective formulated in Section 3 of this research paper:

- **Fulfillment of Objective 1 (Nutritional Transparency):** The platform provides an engaging, responsive digital dining menu featuring 8 verified nutritional metrics per dish, allergen alerts, ingredient listings, and dynamic dietary badges (Vegetarian, Vegan, Non-Vegetarian, Jain-friendly).
- **Fulfillment of Objective 2 (Interactive Meal Customization):** Diners can dynamically modify portion sizes, carb bases, dressings, and protein add-ons, with the client and server recalculating both price and macronutrient totals in real time.
- **Fulfillment of Objective 3 (Contactless Table Ordering):** Physical dining tables are provisioned with dynamic QR tokens, enabling zero-contact onboarding, digital ordering, and instant kitchen ticket generation without waiter intervention.
- **Fulfillment of Objective 4 (Live Kitchen Operations):** The operational Kanban board provides kitchen staff with an intuitive tool to transition orders across 5 verified

stages (Placed, Accepted, Preparing, Ready, Completed), propagating status updates to diner views within 4 seconds.

- **Fulfillment of Objective 5 (Owner Management & Analytics):** Restaurant owners possess full administrative control over menus, categories, ingredients, variants, tables, and customer reviews, supported by real-time analytics on sales and nutritional distribution.
- **Fulfillment of Objective 6 (Data Integrity & Security):** The system enforces robust defense-in-depth security, including PDO prepared statements, output encoding, CSRF tokens, strict multi-tenant isolation, and complete historical snapshot preservation across 17 normalized relational tables.

11.3 Key Architectural and Technological Contributions

The technical execution of Healthy Bite yielded several noteworthy software engineering contributions:

- **High-Performance Lightweight Architecture:** By avoiding heavyweight client frameworks and instead implementing clean Vanilla CSS and modular ES6+ JavaScript, the application achieves rapid sub-second page loads, minimal DOM overhead, and broad cross-device compatibility.
- **Immutable Snapshot Preservation:** The database architecture solves the common industry flaw of retroactive order alteration by taking immutable snapshots of dish names, variant names, customization choices, prices, and macronutrient baselines upon order placement.
- **Secure Multi-Tenant Partitioning:** The platform enforces strict tenant-level relational isolation, ensuring that multiple restaurant branches can co-exist within the same database infrastructure without risk of data cross-contamination or unauthorized access.
- **Operational Transparency:** The live kitchen pipeline bridges the operational gap between back-of-house kitchen preparation and front-of-house diner expectations, reducing perceived waiting times and customer anxiety.

11.4 Future Scope and Research Directions

While the current implementation achieves full functional maturity within its academic scope, several promising extensions have been identified for future research and commercial development. These enhancements are formally designated as Future Work:

- 1. AI-Driven Personalized Dietary Recommendations:** Integrating machine learning models (e.g., collaborative filtering and content-based recommendation algorithms) to analyze past customer dietary choices and recommend personalized meals aligned with specific caloric, keto, diabetic, or athletic goals.
- 2. Automated BMI & Calorie Budgeting Calculator:** Implementing an interactive health profile module allowing users to input height, weight, activity levels, and wellness goals to compute dynamic Basal Metabolic Rates (BMR) and recommended daily caloric thresholds.
- 3. Wearable Device & Fitness Platform Integration:** Integrating health APIs (such as Apple HealthKit, Google Health Connect, and Fitbit Web API) to synchronize consumed meal macros directly with diners' personal fitness trackers.
- 4. Native Mobile Applications:** Developing native mobile applications for Android and iOS using cross-platform frameworks (e.g., Flutter or React Native) to provide offline caching, geofenced table detection, and push notifications.
- 5. Voice-Assisted Conversational Ordering:** Incorporating voice-activated conversational interfaces utilizing Web Speech APIs to facilitate hands-free food ordering and improve accessibility for visually impaired or elderly patrons.
- 6. Commercial Payment Gateway Integration:** Extending the simulated payment layer to integrate licensed commercial payment gateways (such as UPI 2.0, Razorpay, or Stripe), enabling secure automated credit card processing, digital wallets, and automated refunds.
- 7. Multi-Lingual Regional Localization:** Extending language catalogs to support multi-lingual localization across regional Indian and international languages, broadening demographic accessibility.

12.REFERENCES

- PHP Official Manual (php.net/manual) — Used for PDO prepared statements, custom MVC routing, session security, password hashing, and CSRF token generation.
- MySQL 8.0 Reference Manual (dev.mysql.com/doc/refman/8.0/en/) — Used for designing the 16-table relational schema, foreign key cascade rules, and MariaDB database dump exports.
- Mozilla Developer Network (developer.mozilla.org) — Used for modern JavaScript fetch() API calls, localStorage cart state persistence, and DOM event handling.
- OWASP Top Ten (owasp.org) — Used as the security specification for defensive programming (SQL injection defense via parameter binding, XSS prevention via htmlspecialchars, and CSRF protection via hash_equals).
- Bootstrap Icons Documentation (icons.getbootstrap.com) & jsDelivr CDN (jsdelivr.com) — Used for loading all vector interface icons across the application.
- Google Fonts (fonts.google.com) — Used for typography via Inter and Outfit.
- Davidshimjs QRCode.js Repository (github.com/davidshimjs/qrcodejs) — Used for the embedded table QR code canvas engine (public/assets/js/qrcode.min.js).
- Python Tooling Documentation (python-docx, Pillow, openpyxl, ReportLab) — Used for automated generation of the 300 DPI DFD diagrams, project report (.docx), testing spreadsheet (.xlsx), and QA validation report (.pdf).